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A new and straightforward synthesis route for preparing Cds quantum Dots

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ABSTRACT

In this work, highly stable, soluble and luminescent CdS quantum dots (QDots) with a narrow size distribution were synthesized in ethylene glycol using the polyol process and the solvothermal technique. In this case instead of using a conventional highly toxic sulfur source like H_2S, we use elemental sulfur dissolved in ethylene glycol to perform the reaction. When the solvent reaches its boiling point inside the autoclave, sulfur is reduced to S^{-2} and reacts with Cd^{+2} ions to form CdS nanocrystals. Analysis of the spectroscopic and TEM measurements showed that 3 nm monodispersed CdS QDots were synthesized and exhibited high photoluminescence (PL) in the blue green region of the spectra when excited with 355 nm.

1. Introduction

The synthesis and properties of II–VI semiconductor quantum dots (QDots) have been extensively investigated over the last 30 years. The reason for this seems to be their special optical and electronic properties which arise from the quantum confinement of electrons and the large surface area [1–3]. In particular, CdS have been extensively studied due to their potential applications in several technological areas such as solar photovoltaic cells, nano bar codes, field effect transistors, light emitting diodes, photocatalysis and *in vivo* biomedical detection fluorescent tags in biology and the development of chemical and biological sensors [3–6].

Due to the high potential of possible applications displayed by this semiconductor material, a wide range of synthetic routes has been developed in order to obtain monodisperse and more efficient quantum dots. These methodologies allow the size, morphology, size distribution and stabilization of nanoparticles to be controlled.

Among these methodologies, one of the most common routes for preparing CdX QDots is the colloidal chemical one by rapid injection of pyrophoric organometallic reagents into hot coordinating solvents at high temperatures (180–350 °C). However, the starting reagents used in these routes are extremely toxic, pyrophoric, explosive and expensive. These drawbacks prompted several groups to use an alternative route to overcome these problems, such as the preparation of monodisperse CdE (E=S, Se and Te) based on the corresponding organometallic precursors [7], or using a single source precursor to prepare CdS nanocrystals of a small size diameter [8,9]. Other preparatory techniques include sol–gel processing [10], microwave heating [11], photoetching [12] reverse micelle [13], reduction route using $NaBH_4$ [14] solvothermal routers [15–18] and photochemical [19].

In this study, highly luminescent CdS quantum dots (QDots) with a narrow size distribution were synthesized in ethylene glycol using the solvothermal technique, where instead of using a conventional, highly toxic sulfur source like H_2S, or organometallic precursors which are toxic, air and moisture sensitive, we have used elemental sulfur dissolved in organic solvent to react with Cd ions in a Teflon-lined autoclave reactor by a procedure slightly modified to that employed by Rao et al., to synthesize CdS nanocrystals [15], where instead of using tetralin in our method, we use the reducing power of ethylene glycol at boiling point to convert sulfur to S^{-2} which in turn reacts with cadmium ions to yield highly crystalline and soluble CdS nanoparticles. It is worth mentioning here that no capping agent is used to stabilize the particles in solution.

2. Experimental

All the chemicals used were of analytical grade and were used without further purification; cadmium nitrate [$Cd(NO_3)_2$] from the Aldrich company as the Cd source; sulfur from Aldrich dissolved in tetrahydrofuran (THF) and ethylene glycol from Aldrich as well. The schematic illustration of a typical synthesis process of a luminescent CdS QD, is shown in Fig. 1, where different molar ratios of analytical grade [$Cd(NO_3)_2$] and sulfur dissolved in THF were added to a Teflon-lined 70 mL capacity autoclave and placed in an air oven which was preheated to 180 °C for the desired reaction time. Since the reaction is carried out in a closed vessel, the inert condition is not required to prevent sulfur precipitation. After the reaction was complete, the autoclave was taken out of the oven and allowed to cool to room

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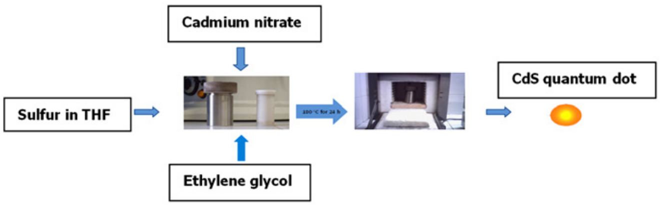

Fig. 1. Schematic illustration of a typical synthesis process of a luminescent CdS QD.

temperature. The reaction mixture presents a yellow color that does not precipitate on centrifugation. Spectroscopic characterization in the UV–visible and infrared region was performed with a Perkin Elmer Lambda 6 spectrophotometer and with a FTIR Bruker IF 66 spectrophotometer, respectively, while room temperature photoluminescence (PL) was measured in an ISS K-2 fluorescence spectrophotometer. The morphology and particle size of the nanoparticles were characterized by the analysis of the transmission electron microscope (TEM) images and selected area electron diffraction (SAED) photographs obtained using a JEOL (JEM3010) transmission electron microscope operating with an accelerating voltage of 100 kV, and by x-ray diffraction analysis using a Rigaku diffractometer model DMAX 2400, using Cu K_α radiation ($l=0.154$ nm) at a scanning speed of $0.1^\circ\,s^{-1}$ in the range of $2\theta=10$–60° with a step 0.02°. The sample for the TEM measurements was dropped onto a carbon-covered 200-mesh copper grid followed by natural evaporation of the solvent, while for the x-ray measurement, since the quantum dots have a very small size distribution and disperse well in ethylene glycol it does not precipitate and for that reason we evaporated the solvent in order to get the x-ray analysis.

3. Results and discussion

The reactions of Cd^{+2} ions with sulfur dissolved in THF in the presence of ethylene glycol within the autoclave reactor at 180 °C resulted in highly luminescent and stable CdS colloids. Fig. 1 shows the blue luminescence when the sample is excited with UV light at 365 nm and the excitation spectra for the samples synthesized as a function of the reaction time. From the excitation spectra we observe a red shift on the absorption peaks from 363nm to 381 nm as the reaction time increases from 24 to 72 h, respectively, meaning that the mean diameter of the nanoparticles is increasing with the reaction time. Also we observe that when the emission peak is set at 550 nm, we are able to see a new excitation peak at 410 nm. Once more, this result indicates that a new quantum dot population grows with the reaction time. On the other hand a comparison with CdS bulk material, which shows maximum absorption around 515 nm [20,21], allows us to say that all of the CdS prepared nanomaterials show an obvious blue shift due to quantum confinement. If the position of the absorption peaks is now compared with the results in the literature [22], we may conclude that the mean size of the CdS nanoparticles can be estimated to be around 2–6 nm.

The formation of CdS nanocrystals can be explained using the reduction mechanism for metallic ions in ethylene glycol proposed by Fievet et al. [23] and Viau et al. [24], which was adapted for our experiment. In the reaction process, ethylene glycol is converted to an aldehyde at the boiling point of the solvent and it reacts with sulfur, reducing it to S^{-2}. Thereafter, it can form in situ H_2S or react directly with Cd^{+2} ions to form CdS nanoparticles as per the mechanism below:

$$2HO{-}CH_2{-}CH_2{-}OH \xrightarrow{\Delta} 2CH_3{-}CHO + 2H_2O \qquad (1)$$

$$2CH_3{-}CHO + S \xrightarrow{\Delta} CH_3CO{-}COCH_3 + S^{-2} + 2H^+ \qquad (2)$$

$$S^{-2} + 2H^+ + Cd^{+2} \rightarrow CdS + 2H^+ \qquad (3)$$

It is interesting to mention here that no capping agent was used in the reaction process. However, the nanoparticles are very

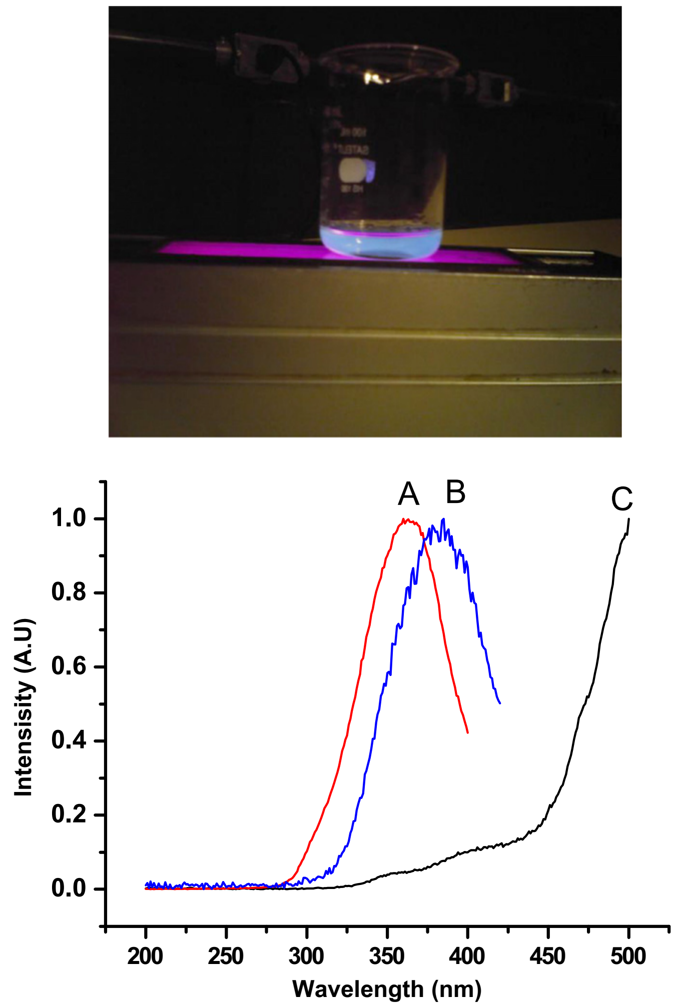

Fig. 2. CdS QDot blue emission and excitation spectra as a function of the reaction time: (A) 24 h and emission peak set at 435 nm, (B) 72 h and emission peak set at 550 nm and (C) 72 h and emission peak set at 550 nm.

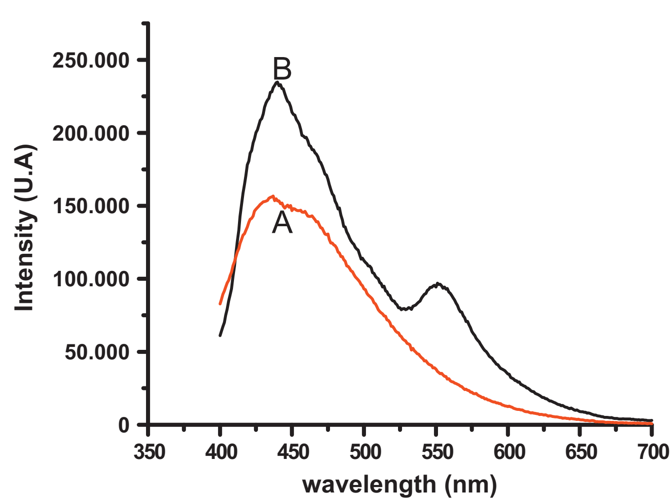

Fig. 3. CdS QDot emission spectra of the sample prepared after (A) 24 h and (B) 72 h.

stable and show no change in the form of their absorption (and emission) and remain dispersed in solution for several weeks without precipitation.

Fig. 2 shows the photoluminecsence (PL) measurement carried out at room temperature for the samples synthesized for (A) 24 h and (B) 72 h when excited at 360 nm. The PL spectrum for the sample synthesized for 24 h presents a strong and broad peak at 435 nm with a tail beyond 700 nm that can be assigned to the recombination of excitons and/or shallowly trapped electron–hole pairs [25,26]. The peak position at 435 nm is a strong indication that this emission is due to very small CdS particles existing in the Q-state [27], which is further confirmed by our TEM results shown in Fig. 3. On the other hand, for the sample synthesized for 72 h, the emission spectra show two peaks: first at the blue region of the spectra at 440 nm and second at 550 nm, where the red shift peak can be attributed to larger particles that grow in the synthesis process, or due to electronic states arising from the surface recombination [28].

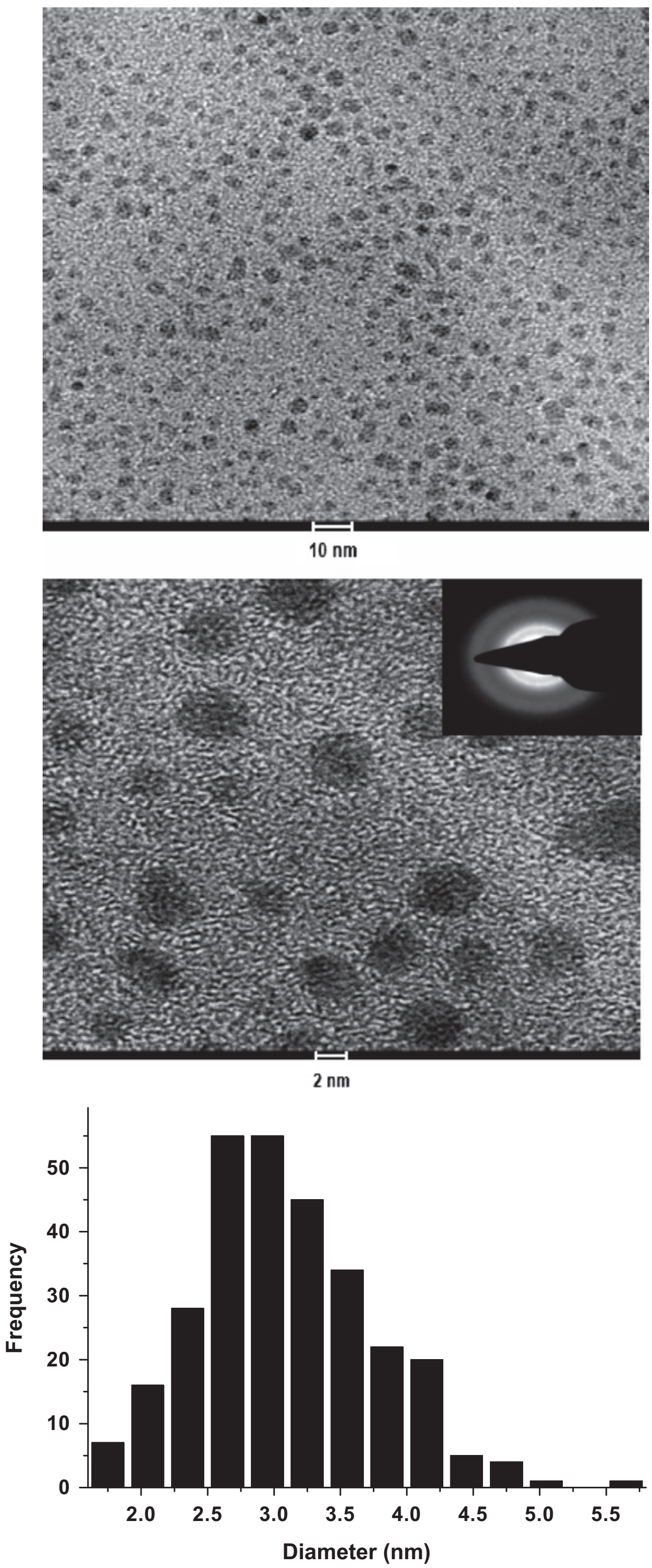

Fig. 4. Images of size distribution and SAED patterns of QDots prepared after 24 h.

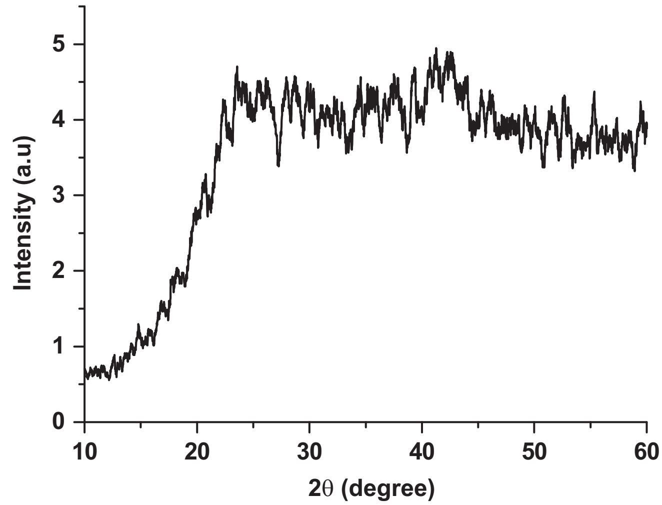

Fig. 5. Powder XRD patterns of CdS QDot nanocrystals prepared after 24 h.

A typical TEM micrograph, size distribution and select area electron diffraction (SAED) patterns for the soluble CdS QDots prepared after 24 h are shown in Fig. 4. These results show that the particles are highly monodispersed and do not aggregate with a nearly spherical shape. The histogram of size distribution was obtained by measuring more than 200 particles from the TEM image. We found the size of particles varies from 2 to 4 nm, with a mean particle size of 3 nm with a standard deviation of about 0.2 nm. The inset of Fig. 3 shows a SAED pattern recorded at an accelerating voltage of 100 kV. The broad and diffuse rings in the electron diffraction pattern are an indication that an extremely small size of the crystallites has been obtained. This result is in agreement with our absorption and emission spectra which strongly indicate that small size particles have been formed. In order to indentify the crystalline structure, Fig. 5 shows the x-ray diffraction patterns for the sample prepared after 24 h, as one can see the combination of a very small amount of sample with the small dimension size of the particles makes the x-ray diffraction patterns be considerably broadened and unresolved. From this result we observe that the XRD pattern of the CdS nanoparticles exhibits three prominent peaks at scattering angles (2θ) of 25, 42 and 52, which could be indexed to scattering from the 111, 220 and 311 planes, respectively, of cubic zinc blend structure, which are in agreement with the Rao et al., results [15]. However due to the patterns broadening we cannot assign indubitably only one crystallographic phase, however from the SAED patterns and the x-ray we can say that the sample is very small and crystalline.

4. Summary

To summarize, we have synthesized highly luminescent CdS QDots with a narrow size distribution using a new synthesis route and a less hazardous cadmium source without any capping agent by employing the solvothermal technique co-assisted by the polyol method at temperatures as low as 180 °C. The as-synthesized CdS QDots are monodisperse and exhibit strong PL in the blue spectral region. The mean particle size of the CdS QDots measured using TEM is 3 nm and the SAED measurement indicates that crystalline and extremely small size particles have been obtained. This value is in good agreement with the estimated size from the UV–visible absorption spectrum and the PL spectrum.

Acknowledgment

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